Varicocele-Associated Infertility and the Role of Oxidative Stress on Sperm DNA Fragmentation.

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Varicocele has been extensively described and studied as the most important reversible cause of male infertility. Its impact on semen parameters, pregnancy rates, and assisted reproductive outcomes have been associated with multifactorial aspects, most of them converging to increase of reactive oxygen species (ROS). More recently, sperm DNA fragmentation has gained significant attention and potential clinical use, although the body of evidence still needs further evolution. The associations between sperm DNA damage and a variety of disorders, including varicocele itself, share common pathways to ROS increase. This mini-review discusses different aspects related to the etiology of ROS and its relation to varicocele and potential mechanisms of DNA damage.

Effect of varicocelectomy on male infertility.

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Varicocele is the most common cause of male infertility and is generally correctable or at least improvable by various surgical and radiologic techniques. Therefore, it seems simple and reasonable that varicocele should be treated in infertile men with varicocele. However, the role of varicocele repair for the treatment of subfertile men has been questioned during the past decades. Although varicocele repair can induce improvement of semen quality, the obvious benefit of spontaneous pregnancy has not been shown through several meta-analyses. Recently, a well-designed randomized clinical trial was introduced, and, subsequently, a novel meta-analysis was published. The results of these studies advocate that varicocele repair be regarded as a standard treatment modality in infertile men with clinical varicocele and abnormal semen parameters, which is also supported by current clinical guidelines. Microsurgical varicocelectomy has been regarded as the gold standard compared to other surgical techniques and radiological management in terms of the recurrence rate and the pregnancy rate. However, none of the methods has been proven through well-designed clinical trials to be superior to the others in the ability to improve fertility. Accordingly, high-quality data from well-designed studies are needed to resolve unanswered questions and update current knowledge. Upcoming trials should be designed to define the best technique and also to define how to select the best candidates who will benefit from varicocele repair.

Varicocele as a progressive lesion: positive effect of varicocele repair.

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Varicoceles are the leading correctable cause of infertility in men who present to an infertility clinic for evaluation. Consequently, the surgical correction of a varicocele, known as a varicocelectomy, is the most commonly performed operation for the treatment of male infertility. The current data suggest that an individual with a varicocele, even with a previously normal semen analysis or documentation of previous fertility, is at risk for subsequent loss of testicular function and infertility. Many of these patients will need to be treated because there is convincing evidence that a varicocele may have a progressive toxic effect on the testes that may ultimately result in irreversible infertility if left untreated. Identifying those individuals with varicoceles that will ultimately cause fertility impairment is still beyond our current clinical capabilities. Current investigative modalities, e.g. semen analysis, testicular measurement, serum gonadotrophin determination, gonadotrophin-releasing hormone (GnRH) stimulation test, and testis biopsy analysis, may be employed to detect early changes in testicular physiology produced by a varicocele.

Lack of significant difference between internal spermatic vein and brachial vein ischemia modified albumin levels in patients with varicocele.

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Varicocele is the most common and surgically correctible cause of male infertility in men attending to infertility clinics. Infertility affects 15% of all couples and male factor is the primary or contributing cause in 40% to 60% of cases. Varicocele has been shown to cause male infertility in about 15% of infertile couples. Molecular mechanisms responsible from varicocele induced testicular dysfunction and male infertility have not been completely unknown. Recent years have witnessed a huge amount of scientific works devoted to the mechanism of varicocele associated male infertility and rapid progress in research on its cellular and molecular mechanisms, including apoptosis and oxidative stress of germ cells. Here we evaluated internal spermatic vein and brachal vein ischemia modified albumin (IMA) level in 40 adult male patients with varicocele. IMA level was analyzed using albumin cobalt-binding test. Spermatic vein and brachial vein IMA levels were 0.381 ± 0.135 ABSU (absorbance units) and 0.385 ± 0.131 ABSU, respectively. There was no statistically significant difference between the two areas. IMA levels in the internal spermatic vein of patients with varicocele should not be used as a marker of hypoxia.

Pathophysiology of varicocele in male infertility in the era of assisted reproductive technology.

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Varicocele is the most common and treatable cause of male infertility. Studies of a rat experimental left varicocele model and human testicular biopsy samples have shown the involvement of various factors in its pathophysiology. Among them, oxidative stress plays a major role in impairing spermatogenesis and sperm function. Therefore, in addition to palpation, scrotal ultrasonography and color Doppler ultrasound, evaluation of testicular oxidative stress (e.g. scrotal temperature is a surrogate parameter) is recommended to enable diagnosis and suitable treatment of varicocele. Varicocelectomy increases the fertilization, pregnancy and live birth rates, indicating improved sperm function; it is therefore important even in couples undergoing intracytoplasmic sperm injection. Routine sperm-function tests are warranted to monitor the sperm quality after varicocelectomy and consequent improvement in the outcomes of assisted reproductive technology. Furthermore, the indications of varicocelectomy in assisted reproductive technology should be widened.

The effects of varicocelectomy on the DNA fragmentation index and other sperm

parameters: a meta-analysis.

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Varicocele is one of the most common causes of reversible male infertility, and 15% of the varicocele patients with normal semen analysis are diagnosed as infertile. According to the current guidelines, varicocelectomy is indicated based on abnormal sperm parameters and not abnormal DNA fragmentation index (DFI) values. Thus, in this study, we performed a meta-analysis of the effects of varicocelectomy on the DFI and other conventional sperm parameters, and determined whether DFI could be used to indicate varicocelectomy for varicocele patients. Through an electronic search of the PubMed, Scopus, EBSCO, and Cochrane databases, we included 7 prospective studies including a total of 289 patients in this meta-analysis. The results showed that varicocelectomy significantly reduced DNA fragmentation (mean difference: -6.86; 95% confidence interval [CI]: -10.04, -3.69; $p < 0.00001$) and improved sperm concentration (mean difference: 9.59; 95% CI: 7.80, 11.38; $p < 0.00001$), progressive motility (mean difference: 8.66; 95% CI: 6.96, 10.36; $p < 0.00001$), and morphology (mean difference: 2.73; 95% CI: 0,65, 4.80; $p = 0.01$). Varicocelectomy reduced DNA fragmentation and improved sperm concentration, progressive motility, and morphology. Additionally, the analysis showed that an abnormal DFI measurement should be considered as an indication for varicocelectomy.

Varicocele management in the era of in vitro fertilization/intracytoplasmic sperm injection.

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Varicocele is the most common surgically treatable cause of male infertility, and often results in alterations in semen parameters, sperm DNA damage, and changes to the seminal milieu. Varicocele repair can result in improvement in these parameters in the majority of men with clinical varicocele; data supporting repair in men with subclinical varicocele are less definitive. In couples seeking fertility using assisted reproductive technologies (ARTs), varicocele repair may offer improvement in semen parameters and sperm health that can increase the likelihood of successful fertilization using techniques such as in vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI), or may decrease the level of ART needed to achieve successful pregnancy. Male infertility is an indicator of general male health, and evaluation of the infertile male with an eye toward future health can facilitate optimal screening and treatment of these men. Furthermore, varicocele may represent a progressive lesion, offering an argument for its repair, although this is currently unclear.

Is varicocelectomy indicated in subfertile men with clinical varicoceles who have asthenospermia or teratospermia and normal sperm density?

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Varicocele is the most common treatable cause of male infertility and is associated with progressive decline in testicular function. Varicocelectomy, a commonly performed operation, is indicated in infertile males with varicoceles who have oligospermia, asthenospermia, teratospermia or a combination of these factors. It is not clear if varicocelectomy is indicated if the patients have normal sperm density associated with asthenospermia or teratospermia. We reviewed 167 patients with varicocele-associated male infertility over a 7-year period (December 1999-November 2005). Pre- and post-varicocelectomy seminal fluid analyses, assessed using the World Health Organization criteria, were obtained at intervals of 4-6 months. Wilcoxon signed rank tests were used to evaluate for statistical significance and P < or = 0.05 was considered significant. The mean age of the patients and their spouses were 35 and 28 years, respectively. The mean duration of infertility was 3.2 years (range, 1.5-7.5). Oligospermia, teratospermia, asthenospermia, oligospermia, asthenospermia and teratospermia (OAT) syndrome and azoospermia were found preoperatively in 106 (63.5%), 58 (34.7%), 154 (92%), 118 (71%) and 15 (9%) patients, respectively. Overall, significant improvements in semen volume (P < 0.001), sperm density (P < 0.001), sperm motility (P < 0.001) and sperm vitality (P < 0.001) were obtained after varicocelectomy. There was, however, no significant improvement in sperm morphology after varicocelectomy (P = 0.220). When patients with preoperative oligospermia (sperm density, or =20 million/mL (normospermia) associated with asthenospermia and/or teratospermia were considered separately, they did not show significant improvement in any of the semen parameters after varicocelectomy (P > 0.05). In addition, azoospermic patients did not show significant improvement in any of the semen parameters (P > 0.05) No significant improvement in semen parameters may be obtained in patients with clinical varicocele and preoperative normospermia. It is possible that only patients with preoperative oligospermia may benefit from varicocelectomy. Larger multi-institutional studies are needed to determine more definitively if asthenospermia or teratospermia in normospermic subfertile males with clinical varicoceles are in fact indications for varicocelectomy.

[A role of varicocele in the development of male infertility and methods of surgical treatment].

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Varicocele is the most common and treatable cause of male infertility. Therefore, surgical treatment of varicocele should be recommended for motivated patients and infertile couples who do not have other identified causes of infertility. Varicocelectomy has been proved to improve sperm parameters and improve chances of successful conception in most patients. However, surgical treatment is associated with certain risks, and therefore, the choice of the optimal treatment is under discussion. A total of 78 articles using a search in MEDLINE database (PubMed) were found and included in the review, dedicated to current concepts of functional anatomy of testicular arteries and veins. The current recommendations of professional communities regarding the selection of patients for varicocelectomy are described. The efficiency and safety of various surgical procedures for varicocele is analyzed. This review suggests high inconsistences in the literature. The available information on the indications for surgical treatment, as well as comparative data on the efficiency and safety of the inguinal, laparoscopic and microsurgical sub-inguinal approaches are presented. When urologist faces with a diagnosis of varicocele, individual approach should be applied, with a discussion of both benefits and possible complications of surgical treatment. Of the many existing techniques, microsurgical ligation of dilated veins is the most preferred.

Elastography to assess the effect of varicoceles on testes: a prospective controlled study.

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Varicoceles are the most common and treatable cause of male infertility. The pathophysiology of varicoceles primarily includes elevated temperature, adrenal hormone reflux, gonadotoxic metabolite reflux, altered testicular blood flow, antisperm antibody formation and oxidative stress. The diagnosis of a varicocele is mainly clinical. However, a Doppler ultrasound is used to obtain clinical data and to more accurately measure testicular size. Acoustic radiation force impulse (ARFI) is an additional technique to simultaneously show different areas with different densities in a colour-coded image and a B-mode or greyscale image. This can be used for structural analysis of testicular tissue and has become an additional method for detecting pathologic tissue alterations. We enrolled 30 patients who had clinically diagnosed with left varicoceles and male infertility (Group 1). All patients were evaluated by history taking, physical examination, a spermiogram and an endocrine profile. Thirty control patients (Group 2) were randomly chosen from patients who had applied to an andrology clinic for infertility; their physical examinations and laboratory results showed normal findings. Mean elastography results were significantly different between the groups, and significantly lower in patients who had varicoceles. The relationship between hormonal profiles and elastography parameters was calculated as statistically significant negative correlations between FSH and elasticity. Additionally, a negative correlation was determined between varicocele grade and elasticity of testes. In conclusion, our prospective study showed that ARFI imaging may be more useful than palpation for determining early damage of testicular structure by varicoceles.